MetaSonic: Advancing Robot Localization With Directional Embedded Acoustic Signals

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***Abstract*—Indoor positioning in environments where GPS cannot be used is a fundamental technology for robot navigation and human-robot interaction. However, existing vision-based localization systems cannot work in low-visibility environments, and existing wireless or acoustic localization systems require specific transceivers, making them expensive and power-intensive — particularly challenging for microrobots. This letter proposes a new metasurface-assisted ultrasound positioning system. The key idea is to use a low-cost passive acoustic metasurface to transfer any speaker into a directional sound source, with the acoustic spectrum varying based on direction. This allows any microrobot with a simple, low-cost microphone to capture such modified sound to identify the direction of the sound source. We develop a lightweight convolutional neural network-based localization algorithm that can be efficiently deployed on low-power microcontrollers. We evaluate our system in a large complex office. It can achieve a direction estimation accuracy of 7.26°, improving by 42.2% compared to systems without the metasurface and matching the performance of a 4-microphone array, with a localization accuracy of 0.35 m.**

***Index Terms*—Localization, robot audition, deep learning, angle estimation, acoustic metasurface.**

I. Introduction

Localization in indoor environments where GPS is unavailable is a crucial challenge for robots, especially microrobots and drones. Accurate indoor localization is essential for enabling precise robot navigation in cluttered environments, efficient cargo handling in warehouses [1], autonomous delivery services [2], and rapid response in disaster rescue scenarios [3]. Most popular vision-based localization methods cannot work in low-visibility environments. Lidar and radar-based systems are expensive and impractical for microrobots [4], [5] with limited sensing capabilities and computational resources. Wireless RSS-based localization systems like BLE and WiFi lack the accuracy needed for precise positioning. While Angle-of-Arrival (AoA) localization systems improve accuracy, they require specialized array transceivers, adding to the complexity, power consumption, and cost.

Received 26 August 2024; accepted 16 December 2024. Date of publication 1 January 2025; date of current version 10 January 2025. This letter was recommended for publication by Associate Editor Yue Wang and Editor Sven Behnke upon evaluation of the reviewers' comments. This letter was supported by the NSF of China under Grant 62473251. *(Corresponding author: Yi Guo.)*

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Digital Object Identifier 10.1109/LRA.2024.3524903

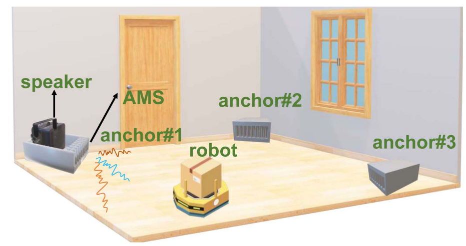


Fig. 1. **MetaSonic Overview.** With the help of a low-cost passive acoustic metasurface, any speaker can broadcast signals with a direction-dependent spectrum. A robot with a basic microphone can localize itself by analyzing this spectrum.

Acoustic-based robot localization has garnered attention [6], [7], [8] due to the lightweight, compact, and pervasive nature of microphones, coupled with the relatively simple processing of audio signals. Traditional acoustic localization relies on the Time-of-Flight (ToF) [9] algorithm to measure distance, but it requires precise synchronization between devices, which is challenging for real-world deployment. Because precise time synchronization requires an extra high-cost network interface with hardware timestamp and involves complex deployment procedure [10]. To solve this issue, research has focused on using microphone arrays mounted on robots [6] to acquire the direction of the sound source. However, microphone arrays require multiple spatially separated microphones, increasing power consumption and device size, making them unsuitable for microrobots. Recently, there has been a growing interest in using passive acoustic metasurface (AMS) [11], [12] on microphones to enhance sound direction sensing ability. AMS consists of multiple passive acoustic unit cells, each capable of modulating the phase or amplitude of sound waves passing through, effectively directing and focusing the sound. It can reduce power consumption but requires integrating metamaterial structures onto the robot microphone, increasing the size of microrobots and deployment complexity.

This letter presents `MetaSonic`, a new acoustic localization system designed to address all limitations of existing approaches, as shown in Fig. 1. Specifically, we designed a passive AMS for the speaker, which alters the acoustic spectrum across different directions, and we refer to this AMS-equipped speaker as an anchor. The AMS consists of multiple acoustic unit cells, each modulating the sound spectrum differently. As a result, the outgoing acoustic wavefront, composed of the combined effects of all unit cells, exhibits a spectrum that varies depending on the direction. When an anchor broadcasts modified sound, any robot with only one basic microphone can capture it and determine its direction to the sound source by analyzing the spectrum.

By collecting the sound from multiple anchors, the robot can easily get absolute localization. This design achieves minimal hardware requirements on both robots and facilities, with the only hardware upgrade being a low-cost metasurface mounted on a basic speaker. The entire system operates within the near-ultrasound range (16–20 kHz), which is virtually inaudible to humans, compatible with most microphones, and insensitive to environmental noise. When short, low-volume near-ultrasound signals are broadcast, they remain barely noticeable. Furthermore, precise synchronization between anchors is not required, as the system relies on direction-based localization.

While the idea of utilizing the passive AMS to create a directional acoustic wave for localization is promising, we need to solve the following challenges to meet the high accuracy requirement of real-world robot navigation:

(1) Designing the optimal AMS to maximize the localization accuracy: The AoA accuracy is highly related to the design of the passive AMS. There are many possible designs for the passive structure, and each will encode different spatial features, thus affecting the performance of AoA estimation. Therefore, the optimal design of the AMS structure should maximize the difference of spatial encoding features at different frequencies, which can be used to improve localization accuracy. We need to build an optimization model to minimize the average similarity of encoding features across all possible angles and the maximum similarity of encoding features between different angles to search for the optimal design of the passive structure.

(2) Developing a lightweight localization algorithm that can maximize accuracy while being deployable on microrobots: The relationship between the acoustic spectrum and direction is highly complex. Additionally, sampling noise and environmental noise can interfere with localization. Traditional convex optimization models struggle to handle this complexity. Therefore, we need a lightweight algorithm that can efficiently and effectively extract direction information from the noisy spectrum and can be deployed on computation-constraint devices.

(3) How to improve the system scalability?: In large environments with multiple rooms, multiple anchors are required to work simultaneously to extend the system coverage. How do we embed the sound with unique IDs so that these anchors can be identified by analyzing the sound? How do we deploy the anchors to maximize the coverage? These considerations form the foundation of `MetaSonic` 's design, aiming to create a low-cost and scalable acoustic localization system.

To address these challenges, we refine `MetaSonic` and make following contributions:

- Optimal acoustic metasurface design: We build a theoretical model for AMS and develop an optimization algorithm to identify the optimal configuration, achieving the best directional diversity on the spectrum.
- Lightweight neural network-based acoustic localization: We develop a tiny convolutional neural network to extract spatial features from the acoustic spectrum and predict location. Additionally, we deployed this neural network model onto a low-power microcontroller (MCU) to validate its feasibility for use in microrobots.
- Signal encoding and deployment strategy: To improve the system scalability, we first design an acoustic signal encoding protocol to identify each anchor. Then the robot can decode the anchor ID from the modulated sound. We further propose a deployment strategy for large-scale environments and a neural network switching strategy to update the network parameters when receiving multiple anchors' sounds.

Summary of results: We evaluated our system in a complex office of approximately 100 m^2, consisting of two rooms and a lobby, with 12 anchors deployed throughout the environment. Each anchor consists of an off-the-shelf speaker with a passive AMS. The results demonstrate that our AMS can improve AoA estimation accuracy by 42.2% compared to systems without the AMS. The mean AoA localization error is 7.26° and the 2D localization error using four anchors is 0.35 m. Our system can be deployed on an MCU with a power consumption of 80 mW.

II. Related Works

We review the related works regarding robot indoor localization and acoustic metamaterials.

A. Robot Indoor Localization

Sound-based robot localization systems, which use reference landmarks (natural walls [13] or artificial acoustic anchors [9]) for positioning, have garnered a rich interest. We categorize past works into two types: *(1) Time of Arrival (ToA):* The study [13] uses ToA from wall reflections and a 3D map to calculate its position. This method requires prior modeling of all walls, making it unsuitable for complex environments. The study [9] places multiple speakers as anchors and uses microphones on the robot to determine its position by calculating the arrival times between anchors and microphones. This approach requires precise time synchronization between speakers and the robot, increasing cost and deployment complexity. *(2) Direction-based localization:* The study [6] measures the robot's direction relative to multiple anchors for positioning. AIM [7] places microphone arrays on anchors to determine the AoA of drone propeller sounds for positioning. While AoA methods avoid time synchronization issues, they typically require multiple microphone arrays. Unlike them, our system uses passive acoustic metamaterial structures to create directional acoustic fields, allowing robot self-localization with a single microphone without time synchronization issues. Besides acoustic methods, numerous localization methods have been proposed. RF-based localization [14], [15] requires additional receivers or transmitters to be installed on the robot. Inertial measurement methods [16] have cumulative errors and cannot work in long-term tracking, and vision-based methods [17] are ineffective in low-visibility conditions, such as those caused by dust, fog, or smoke. Hence, acoustic localization systems offer low energy consumption and robustness to lighting conditions.

B. Acoustic Metasurface

AMS is a powerful tool for controlling acoustic waves, enabling applications such as localization [11], imaging [18], and communication [19]. Various structures have been proposed, such as coiled-up structures [20] and Helmholtz resonators [21]. Our system adopts the coiled-up structure [20] because it is well-parameterized and easy to produce via 3D printing. AMS enhances the sensing performance of microphones or speakers. The study [19] optimized the AMS units' configuration to enhance the beamforming capability of a speaker array. In contrast, our research focuses on using AMS units' configuration to enhance the spatial diversity of sound. MetaAng [12] uses AMS to cover microphones for AoA estimation. Owlet [11] employs designed acoustic metamaterial structures on microphones for AoA estimation. Although these methods can estimate incident directions, they require AMS to be placed on microphones, which increases the robot's hardware burden and are limited to low-accuracy 1D localization. Unlike past works, our approach designs a specific AMS structure on the speaker side to allow the anchor to generate acoustic waves with distinct spectral characteristics in different directions, enabling the robot to achieve 2D localization with just a single microphone.

III. System Architecture

Our design philosophy is to achieve high localization performance with minimal modifications to existing systems. With smart speakers becoming increasingly ubiquitous, our approach only requires adding a passive AMS to these speakers as shown in Fig. 1. This simple addition allows the speaker to create direction-dependent sound features without altering the original hardware. On the robot side, no modifications to the hardware are required—only a software upgrade. The system design can be divided into two main components:

- **Acoustic Anchor**: On the hardware side, each anchor consists of a basic speaker paired with a 3D-printed passive AMS. The AMS modulates the speaker's sound wavefront to generate sound signals with distinct spectral features in different directions. We design the passive AMS structure and find its optimal configuration. (see Section IV-B) On the software side, the broadcast signal is designed to encode the unique ID of the anchor. We design a chirp signal as a preamble to embed spatial features and an encoded field to embed the anchor's ID. (see Section IV-D)
- **Robot Localization Algorithm**: The robot uses its built-in basic microphone to receive sound signals from the speakers. It first processes them to extract the anchor IDs. Then, the robot can accurately determine its direction relative to every anchor using a deep neural network-based localization algorithm (see Section. IV-C) and achieve precise absolute localization by aggregating data from multiple anchors.

IV. System Design

In this section, we detail our system design. First, we explain the theoretical model of the passive AMS. Next, we introduce the optimization algorithm for the AMS configuration to maximize spatial diversity and localization accuracy. Following that, we introduce our localization algorithm and discuss how to extend our system to large scenarios.

A. Boosting Speaker Directional Diversity With AMS

We firstly present the wave transmission model of the AMS to explain how it enhances the speaker's ability to generate signals with a direction-dependent spectrum.

■ **AMS Unit Cell**. Before deriving the sound transmission model of the AMS, we first introduce the AMS unit cell. Our AMS structure comprises an array of unit cells proposed in [20], each capable of modulating the phase and amplitude of the input sound wave by adjusting its internal air tunnel. Fig. 2(a) shows the internal structure of a unit cell which contains a coiled-up structure. Assuming the sound wave passes through the unit cell from the left, the coiled-up path in each unit cell adjusts the travel time of the input sound wave, thereby altering the output

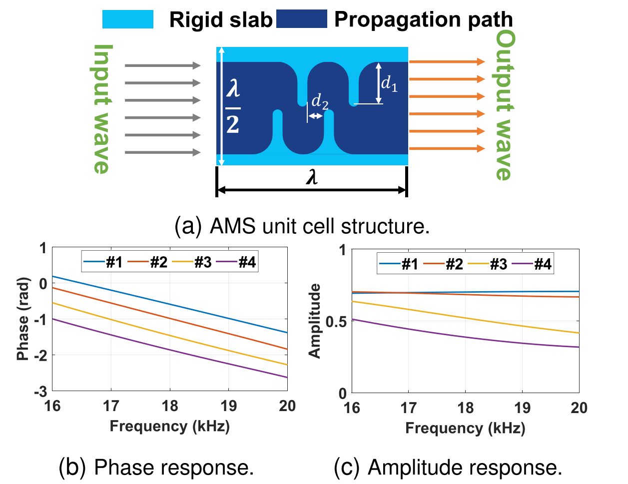


Fig. 2. **AMS Unit Cell.** (a) shows the structure of the AMS unit cell, while (b) and (c) are the phase response and Amplitude response of four unit cell types.

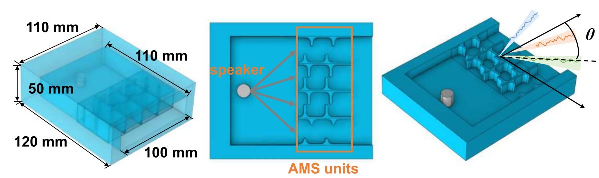
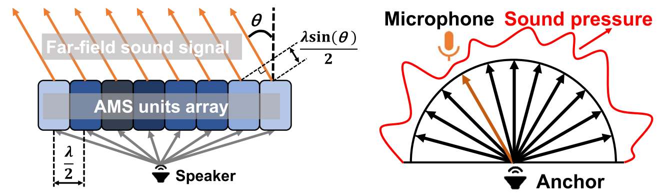


Fig. 3. Structure of AMS array.

wave's phase delay and amplitude attenuation. The external size of the unit cell is proportional to the wavelength λ of the center frequency sound. The phase delay and amplitude attenuation are determined by the parameters d_1 and d_2. By tuning the parameters d_1 and d_2, we identify 16 distinct unit cell types. For each unit cell, we can obtain the phase delay and amplitude attenuation across the full frequency spectrum using the COMSOL simulator, a widely recognized finite element-based multiphysics simulation tool. Fig. 2(b) and (c) illustrate four example unit cell types with the parameters (d_1, d_2) in mm as follows: #1: $(1.24, 4.32)$, #2: $(1.84, 2.24)$, #3: $(2.24, 4.14)$, and #4: $(2.64, 3.78)$. The cell size is determined using $\lambda = 20$ mm (namely, setting the operating frequency as 17 kHz). As the frequency increases, the phase increases for all unit cells. However, each cell exhibits different initial phase offsets and a unique attenuation pattern across the spectrum. By combining an array of these units, we can precisely control the sound spectrum in each direction.

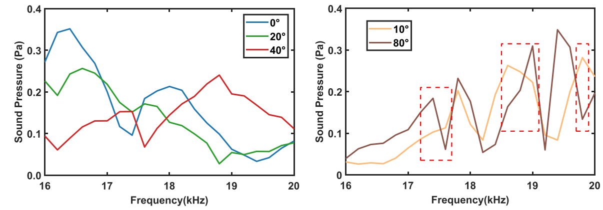
■ **AMS Array**. The passive AMS is configured as a rectangular structure with an internal cavity, as shown in Fig. 3. A speaker is centrally placed within the cavity and is directly connected to the AMS unit cell array. The speaker emits sound waves that are first modulated by the array of unit cells. These modulated waves are then transmitted to the external environment. The outgoing waves are embedded with the direction-dependent features created by the AMS array.

■ **Array Sound Propogation Model**. Consider there are N unit cells in a 1D array in front of a speaker as Fig. 4(a). The emitted sound first propagates over the air, then passes through the unit cells, and finally, all waves of unit cells combinate to generate the far-field sound signal. Assume the emitted sound



(a) Far-field transmission. (b) Direction diversity.

Fig. 4. **Transmission Model.** (a) shows the sound wave passes the AMS unit cell array and generates the far-field signal. (b) shows the sound pressure diversity across directions.



(a) Spectra in three directions. (b) Spectral differences.

Fig. 5. **Amplitude spectrum Diversity.** (a) shows spectra in three directions of an AMS array with 10 unit cells. (b) shows the spectral differences when the cosine similarity is high.

wave at frequency f (its wavelength is λ) has a complex sound pressure of $P_0(f)$. Let d_i be the distance between the speaker and the i-th unit, and $a(d_i)$ be the amplitude attenuation over the distance d_i. The sound pressure at i-th unit cell is:

$$P_{i0}(f) = P_0(f)a(d_i)\exp\left(j\frac{2\pi d_i}{\lambda}\right) \tag{1}$$

Then, when the acoustic wave passes through the unit cell, there will be a phase delay $\phi_i(f)$ and amplitude attenuation $b_i(f)$ at frequency f, which can be determined via simulation. The output sound pressure of the i-th unit cell is:

$$\begin{aligned} P_i(f) &= P_{i0}(f)b_i(f)\exp(j\phi_i(f)) \\ &= P_0(f)a(d_i)b_i(f)\exp\left(j\left(\phi_i(f)+\frac{2\pi d_i}{\lambda}\right)\right) \end{aligned} \tag{2}$$

By summing the effects of N unit cells, each spaced $\frac{\lambda}{2}$ apart, and considering the far-field sound transmission model, the complex sound pressure at angle θ and frequency f is:

$$\begin{aligned} P(\theta, f) &= \sum_{i=1}^{N} P_i(f)\exp\left(j\frac{2\pi f(i-1)\frac{\lambda}{2}\sin(\theta)}{c}\right) \\ &= \sum_{i=1}^{N} P_0(f)a(d_i)b_i(f) \\ &\quad \times\exp\left(j\left(\phi_i(f)+\frac{2\pi d_i}{\lambda}+\frac{2\pi f(i-1)\frac{\lambda}{2}\sin(\theta)}{c}\right)\right) \end{aligned} \tag{3}$$

where c is the sound speed in air. $\frac{2\pi f(i-1)\frac{\lambda}{2}\sin(\theta)}{c}$ is the phase difference between the i-th unit cell and the first unit cell in the array caused by the transmission distance difference in the far-field. From (3), we observe that for a given frequency, the sound pressure varies across different directions, as illustrated in Fig. 4(b). By analyzing the sound pressure, the receiver can estimate the direction of the incoming sound wave. However, relying on sound pressure diversity at one frequency is inaccurate in direction estimation, as some directions may exhibit similar sound pressures. To enhance the directional diversity, we can use wideband signals as the input wave, resulting in distinct amplitude spectra in various directions, which we refer to as spectrum diversity. Fig. 5(a) demonstrates the variation in amplitude spectra across different directions of AMS in simulation, allowing us to determine the direction by identifying the one with the highest similarity. Clearly, the accuracy of localization depends significantly on the degree of spectrum diversity.

B. Optimal AMS Structure Design

As previously discussed, the passive AMS can enhance the spectral diversity of a speaker. How can we determine the optimal configuration of the passive AMS array to maximize spectrum diversity and thereby enhance localization accuracy? Specifically, an optimization algorithm is required to determine the optimal combination of the passive structure array (i.e., selecting the most suitable unit cell type from the 16 available types for each cell) to maximize directional diversity. We first quantify this spectral similarity $G_{i,k}$ using cosine similarity, defined as follows:

$$G_{i,k} = \frac{v_i \cdot v_j}{\|v_i\|\|v_j\|} \tag{4}$$

where v_i and v_j are the vectors representing the sound spectra at directions θ_i and θ_j, respectively. Each vector is composed of the spectral power profile across bands from f_1 to f_m, expressed as $v_i = \{P(\theta_i, f_1), \ldots, P(\theta_i, f_m)\}$. The spectral power $P(\theta_i, f_m)$ can be computed using (3). A cosine similarity value close to 1 indicates a maximum similarity, while 0 indicates a minimum similarity. To achieve the best localization accuracy, we need to minimize the similarity across any two different directions. We formulate the optimization problem as $\min \sum_{i\neq k} G_{i,j}$. Additionally, we introduce $G_{\max}$, the maximum value among all $G_{i,k}$, to prevent any two directions from being excessively similar. The optimized variable $X = (x_1, x_2, \ldots, x_n)$ represents the selection of AMS unit types, where $x_i \in \{1, 2, \ldots, 16\}$ indicates that the i-th position is assigned to type x_i from the 16 available AMS unit types. . The ultimate optimization problem is given by:

$$\min_{(x_1,x_2,\ldots\ldots,x_n)} \sum_{i\neq k} G_{i,j} + \beta G_{\max} \tag{5}$$

where β is the weighting factor for the loss term. In practice, we consider the optimization of an array composed of 20 unit cells. The angular domain is discretized into 90 directions, each with a separation of 1°, while the frequency domain is discretized into 40 frequencies ranging from 16 kHz to 20 kHz. The unit cells are selected from 16 types identified in prior analysis. The amplitude spectra for the 90 directions are calculated according to (5). The optimization problem is then solved using the Adam optimizer in PyTorch. Fig. 6 presents the similarity distribution map of the random and optimal combinations. We observe that the random combination exhibits large areas of high similarity, whereas the optimal combination demonstrates a significant reduction in these regions after optimization. This trend is corroborated by the average cosine similarities of 0.82 and 0.7, respectively. Notably, this cosine similarity metric is intended for AMS

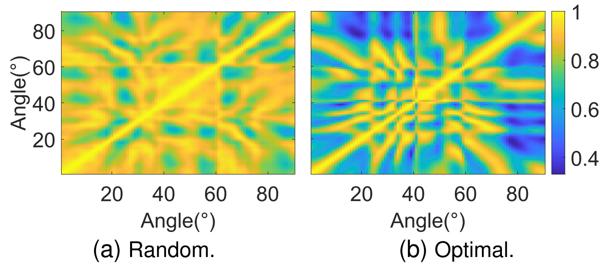


Fig. 6. **Similarity Distribution Map.** (a) shows the cosine similarity among directions with random combination and (b) with optimal combination. The diagonal elements display high values due to self-similarity. Off-diagonal values illustrate the similarity between different directions.

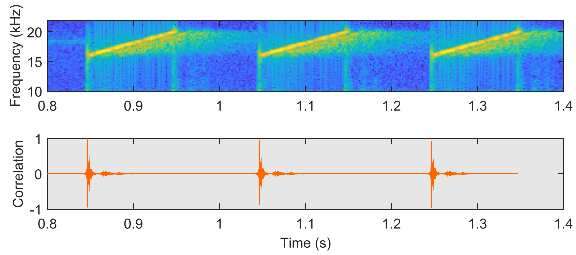


Fig. 7. **Chirps and Correlation.** The top picture shows the periodic chirps' spectrum via short-time Fourier Transform (STFT); the bottom picture shows the correlation result in the time domain, and the peak time is the start of the chirp.

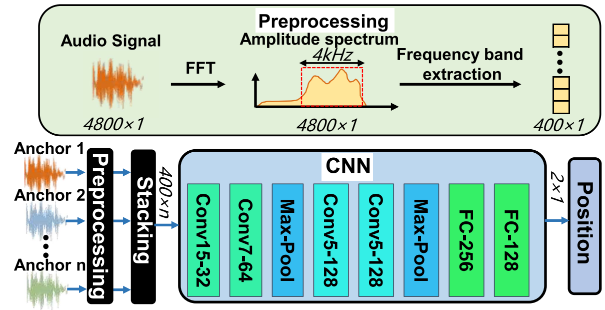


Fig. 8. **CNN Architecture.** The top picture shows the preprocessing of the received signal and the bottom picture provides a detailed view of the CNN architecture. The input consists of signals from n anchors, and the output is the estimated position (x, y).

configuration optimization rather than direct directional differentiation. It may overlook some local differences and be further enhanced with fine-grained feature extractors. For example, as shown in Fig. 5(b), even when the cosine similarity between 10° and 80° exceeds 0.95, their spectra exhibit significant local differences, enabling effective differentiation.

C. Probing and Localization

Next, we introduce the localization sound we broadcast and how to calculate the position by analyzing the received sound.

■ **Broadcasting Chirps in Anchors**. To ensure accurate detection and precise localization, it is crucial to select an appropriate signal for broadcasting. We choose a wideband chirp signal due to its excellent autocorrelation property, enabling easy detection via correlation. The chirp is a sinusoidal waveform with a frequency that linearly increases or decreases over time. The discrete-time chirp $s[n]$ is defined as:

$$s[n] = \sin\left(2\pi\left(f_0 + \frac{1}{2}k\frac{n}{f_s}\right)\frac{n}{f_s}\right) \quad (6)$$

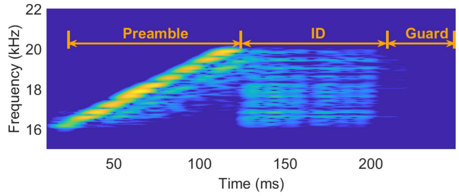
where f_0 is the starting frequency, k is the frequency sweep rate, and f_s is the sampling frequency. Fig. 7 depicts the received chirps and the correlation results.

■ **Deep Neural Network for Localization**. To extract spatial features from the received chirp signal for localization, we utilize a simple Convolutional Neural Network (CNN) model, which has proven effective in other localization tasks. We use CNN because it can extract the spectral feature and differentiate two similar spectra based on its local waveform feature. Fig. 8 shows the architecture of the N -anchor system's CNN model. The raw probing sound for each anchor is 100 ms in duration and sampled at 48 kHz, resulting in 4800 data points. First, we transform the raw time-domain signals into their spectra using FFT over the entire spectrum (24 kHz in our settings). We then extract the relevant frequency band of 16–20 kHz, yielding 400 data points per anchor. These data are stacked into a $400 \times N$ matrix. Subsequently, the network, comprising 4 convolutional layers and 2 fully connected layers, outputs the 2D coordinates $p = (x, y)$. The network is lightweight and we can deploy it on low-cost edge devices. The network is trained using the Mean-Squared Loss function of $Ł = \frac{1}{N}\sum_{i=1}^{N}(p_i - \hat{p}_i)^2$ where p_i is the predicted position and $\hat{p}_i$ is the label, and N is the batch size. When there is only one anchor, the input is 1D sound and the output is the 1-D direction θ to the anchor. When there are multiple anchors ($N \geq 2$), the stacking input of multiple anchors sounds combines the spatial features from multiple anchors, and then CNN will analyze them globally, directly outputting the 2D coordinates. This integrated approach enhances accuracy compared to methods that analyze features independently.

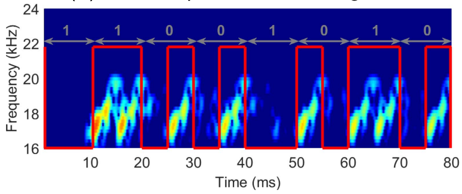
D. Scalability in Large Scenarios

To realize the scalability of `MetaSonic` in large-scale scenarios. First, we introduce the encoding method to distinguish multiple anchors. Then, we discuss anchor deployment strategies and the network switching mechanism.

■ **Time Division Multiple Access**. When deploying multiple anchors, distinguishing between them is crucial. Existing studies use frequency division multiple access (FDMA) or time division multiple access (TDMA) to encode the anchor signals. The FDMA divides the operating frequency band into multiple sub-bands and assigns them to different anchors. Anchors can be differentiated via frequency bands. However, this approach is unscalable when there are numerous anchors because the acoustic band is limited. Therefore, we use the TDMA method, which divides time into multiple slots, with each anchor occupying a unique slot. In practice, all anchors will roughly synchronize themselves using the network synchronization service and then broadcast their signals in a predefined time order and slots to prevent interference. Each signal contains three fields: Preamble, ID, and Guard Interval. The preamble is a 100 ms chirp signal used by the receiver for localization and alignment with the ID field. The ID field contains 8 bits, with each bit lasting 10 ms. These bits are encoded using FM0, with the final bit reserved for parity checking, as illustrated in Fig. 9(b). This



(a) Receive periodic sound signal.



(b) ID encode.

Fig. 9. **Decoding Acoustic ID.** (a) shows the received signals, which comprise three parts: preamble, ID, and guard interval. (b) shows the ID section encoded using FM0, with the red line representing the waveform of the decoded signal.

encoding scheme supports up to 128 anchors. A 100 ms guard interval is included to prevent interference from echoes due to synchronization errors and sound delay.

■ **Deployment and Network Switching Strategy**. The deployment is based on the anchor's optimal coverage area, which is a sector with a 90° angle range and radius of r. The angle range is determined by the anchor's configuration optimization. The optimal distance r is determined by the Signal-to-Noise Ratio (SNR). Based on our experience, the signal's SNR decreases as distance increases and when the SNR falls below 15 dB, the signal detection accuracy degrades which may increase the localization error. Specifically in our system, the r is 6 m. It is worth noting that beyond optimal coverage area does not mean the signal is unusable, it simply means the performance is not optimal. Based on the optimal coverage area, anchors are deployed at intervals of r, and their orientations are adjusted to ensure that a larger area falls within the optimal coverage of at least two anchors. When multiple anchors are deployed, not all locations can be covered by all anchors' signals, and this is an issue for our localization algorithm. How can we address the neural network's varying input channels? To solve this issue in large scenarios, we adopt the following strategy: We identify the IDs of the two strongest received signals and then match them to the corresponding 2-anchor network for prediction. Due to the deployment locations of the anchors, not every combination of two anchors requires a trained network. Therefore, the actual number of networks needed is far less than $\binom{n}{2}$.

V. Implementation

In this section, we present the implementation of the `MetaSonic` . Fig. 10 illustrates the system's hardware setup, comprising an anchor node and a receiver node. The software code, hardware design, and environment configuration are available on the project website https://anplus.github.io/metasonic/.

Anchor: Each anchor node includes a passive AMS and a Bluetooth speaker (JBL Go Essential) with a power consumption of 3.1 W. Specifically, the AMS with 10 unit cells is configured through an optimization algorithm. Each unit cell's size is $10\,\text{mm} \times 10\,\text{mm} \times 20\,\text{mm}$ determined by the sound frequency of 17 kHz. The system broadcasts signals ranging from 16 kHz to 20 kHz. Additionally, the AMS is fabricated from PLA material using a JGaurora Z603s 3D printer. A laptop is connected to the speakers via Bluetooth 4.2 to generate the broadcasted signals, as depicted in Fig. 9. The cost for the AMS is about 15 USD. **Receiver**: The receiver node consists of four MEMS microphones (MSM261S4030HOR), each with a power consumption of 1.35 mW and a sampling rate of 48 kHz. Furthermore, the receiver node includes an AGV (Eaibot smart). The AGV is equipped with dual LiDAR and IMU sensors for SLAM (Simultaneous Localization and Mapping). The AGV employs the ROS gmapping SLAM framework to generate a 2D grid map with an average localization accuracy 5–10 cm.

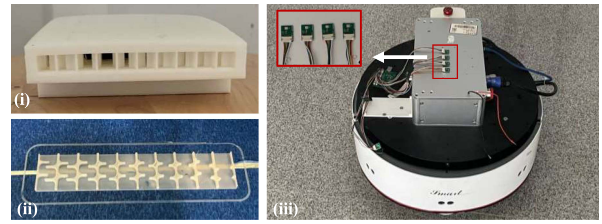


Fig. 10. **Implementation.** (i) the passive AMS. (ii) the 3D printing of the AMS unit cell array. (iii) the AGV platform and the MEMS microphone array.

Training Settings: We utilize AGV-mounted microphones and AGV's localization system to collect datasets and record ground truth data. The chirps emitted from four deployment anchors, with frequency ranges of 16–20 kHz, 18–20 kHz, and 19–20 kHz (corresponding to bandwidths of 4 kHz, 2 kHz, and 1 kHz, respectively), are collected in each sample. The dataset of 5000 samples is uniformly distributed in three scenes: a $3.5\,\text{m} \times 4\,\text{m}$ small room, a $2\,\text{m} \times 14\,\text{m}$ lobby, and a $5\,\text{m} \times 12\,\text{m}$ large room. The dataset is divided into training, validation, and testing sets with a ratio of 7:1:1, and three different models are trained for each scene. The networks are trained using the ADAM optimizer with a learning rate of 0.0018 over 1000 epochs. The network configuration is detailed in Fig. 8. Training is conducted on an NVIDIA GeForce RTX 4060 GPU, completed in under 15 minutes for a four-anchor network. Time consumption for one-time prediction is 0.026 s for a four-anchor network. We also try to compress our model from 17 MB to 670 KB and deploy it on edge devices like microrobots, achieving a 320 ms inference time, 80 mW power consumption, and only a 10% accuracy drop, demonstrating its feasibility for low-cost microrobots.

VI. Evaluation

In this section, we conduct extensive experiments to evaluate the practical performance of the `MetaSonic` . We deploy 12 anchors in an office that is filled with the noise of various machinery, along with tables and chairs, creating a practical operational setting.

A. *Validation of the Effectiveness of Passive AMS*

We start by evaluating the effectiveness of passive AMS by testing the AoA estimation accuracy of the one-anchor system. Two datasets are collected in the small room: one with AMS and the other without AMS. The two datasets have 1200 samples at the same points. We fix the anchor's position at (x_o, y_0) and the angle ground truth is calculated by $\theta = \text{atan2}(\frac{y-y_o}{x-x_o})$ where

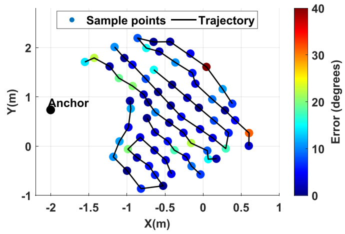


Fig. 11. With AMS.

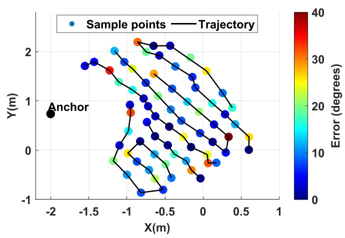


Fig. 12. W/o AMS.

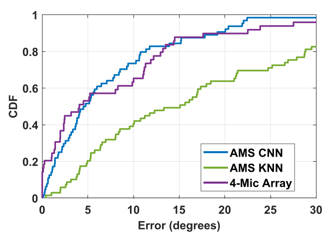


Fig. 13. CDF of AoA error.

(x, y) is the each point's coordinate obtained from the AGV 2D SLAM system which is detail in the Section V. We use a one-anchor CNN and set the angle as output to train the AoA estimation model. We select 64 evenly distributed trajectory samples within the angle range of 90° to evaluate the AoA estimation error. As shown in Figs. 11 and 12, the system with the AMS shows significantly lower error compared to the system without the AMS, with mean AoA estimation errors of 7.26° and 12.56°, respectively. The percentage reduction in error is approximately 42.2%. The enhancement in estimation accuracy can be attributed to the increased directional diversity introduced by our AMS. Also, errors are randomly distributed along the trajectory, likely resulting from environmental and sensor noise.

B. Performance Comparison of Other AoA Methods

We evaluate the K-Nearest Neighbors (KNN) method and the 4-microphone array GCC-PHAT-based method as the baselines. For the KNN method, we evaluate the same samples in the small room as CNN. The KNN parameter is set to $k = 5$, using 90% of the dataset for training, with Euclidean distance as the metric. As shown in Fig. 13, the KNN method produces an error of 17.27° which is significantly higher than the 7.26° achieved with the CNN method. This demonstrates that KNN's simplicity limits its ability to extract shape features from the spectrum as effectively as CNN. For the 4-microphone array method, we utilized the same microphones as those in the CNN setup, spaced 10 cm apart. While keeping the anchor fixed, the microphone array collects data from 60 evenly distributed points, as shown in Fig. 13. The mean AoA error of 4-microphone array is 7.76°, which closely matches the 7.26° achieved by the CNN model. This demonstrates that our method delivers comparable performance to the 4-microphone array while significantly reducing power consumption, cost, and size.

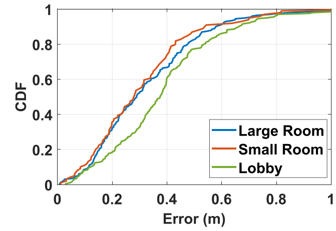


Fig. 14. Impact of scenarios.

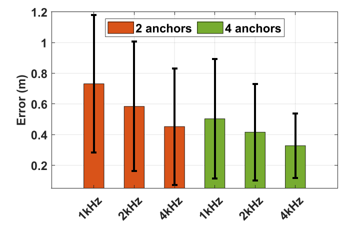


Fig. 15. Bandwidths.

C. Overall Performance in Different Scenes

We evaluate the overall performance of our system based on the dataset in three types of scenes: the small room, the lobby, and the large room. We deploy 4 anchors in each scene. We train three distinct CNN models, each tailored to a specific scene. The input for each CNN corresponds to data from four anchors, with the output representing the estimated position. Fig. 14 indicates that our system achieves a mean error of 0.31 m, 0.33 m, and 0.39 m in the small room, the large room, and the lobby, respectively. The reason for this difference in error rates is the variation of coverage rates. The coverage rate is defined as the proportion of the area where the optimal coverage areas of at least two anchors overlap relative to the total area. The rates for the small room, large room, and lobby are 99%, 92%, and 78%, respectively. Thus, the high coverage rates (over 90%) in the small and large rooms correspond to lower localization errors, while the lower rate in the lobby results in higher localization errors.

D. Impact of Bandwidth

We evaluate the localization accuracy as a function of bandwidth using the dataset collected in the large room. The chirp frequencies are set to 16–20 kHz, 18–20 kHz, and 19–20 kHz, corresponding to bandwidths of 4 kHz, 2 kHz, and 1 kHz, respectively. Fig. 15 illustrates the errors across different bandwidths. The results show that errors decrease as bandwidth increases in both two-anchor and four-anchor systems. For instance, in the two-anchor network, the mean errors are 0.78 m, 0.58 m, and 0.5 m for bandwidths of 1 kHz, 2 kHz, and 4 kHz, respectively. Compared to 1 kHz, errors are reduced by 25.6%

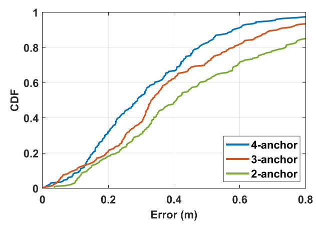


Fig. 16. Impact of anchors.

TABLE I
SWITCHING STRATEGY IN THE LARGE ROOM AND LOBBY

	The large room				The lobby			
Anchor ID	(1,2)	(1,3)	(2,3)	(3,4)	(5,6)	(5,7)	(6,7)	(7,8)
Coverage Rate (%)	21.5	8.2	10.0	20.3	12.0	8.5	9.7	9.7
Mean Error (m)	0.40	0.32	0.34	0.36	0.4	0.46	0.34	0.43
Total Mean Error (m)	0.38							

at 2 kHz and 35.9% at 4 kHz. This improvement is attributed to the wider bandwidth, which allows the system to embed higher-dimensional spectral features, referred to as fingerprint features.

E. Impact of Number of Anchors

To investigate the impact of hardware deployment, we evaluate localization accuracy as a function of the number of anchors using data collected in the small room. As shown in Fig. 16, the mean errors are 0.5 m, 0.4 m, and 0.33 m for the two-anchor, three-anchor, and four-anchor systems, respectively. Compared to the two-anchor system, errors are reduced by 20% with three anchors and 34% with four anchors. Using the coverage rate defined in Section VI-D, the coverage rates are 51%, 65%, and 92%, respectively. These findings demonstrate a significant improvement in performance with additional anchors, primarily due to enhanced signal coverage.

F. Validation of Switching Strategy

We validate our switching strategy in a scenario consisting of a large room and a lobby, covering approximately 90 m^2, with 8 deployed anchors. A dataset of 4,000 samples is collected and divided into 8 groups based on the two strongest signal IDs for each sample. The coverage rate, representing the proportion of data in each group, is calculated and summarized in Table I. Each group is then used to train a corresponding 2-anchor network. During testing, samples are mapped to their respective prediction networks based on the two strongest signal IDs. The mean error for each network and the overall mean error are shown in Table I. The strategy achieves a localization error of 0.38 m, demonstrating high accuracy with a manageable number of networks.

VII. CONCLUSION

We present `MetaSonic`, an innovative acoustic localization system that provides precise indoor positioning for various robotic applications using anchors with passive AMS and a single microphone. This cost-effective, low-power solution simplifies deployment by serving as an add-on to smart speakers and addresses privacy concerns associated with vision-based systems. Key contributions include a 3D-printed metamaterial structure, the elimination of complex sensors, AI integration for efficient position computation, and scalability in large scenarios. `MetaSonic` offers a robust and versatile localization solution, paving the way for advanced robotic tasks in complex environments.

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